PAULA FUDOLIG

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Research Interests

Multi-wavelength studies of black hole – galaxy co-evolution.

Education

George Mason University Aug. 2023 – Present
Doctor of Philosophy in Physics *Fairfax, VA*

Boston University Aug. 2019 – May 2023
Bachelor of Arts in Astronomy and Physics *Boston, MA*

Project Experience

Radio Emission in Radio-Quiet Quasars May 2025 – Present
Graduate research, in collaboration with the National Radio Astronomy Observatory *Fairfax, VA*

- VLBA data reduction in **AIPS**, imaging, and analysis of optically-selected quasar observations at 6 GHz to trace the origin of observed radio-quiet emission.

Varstrometry for Dual AGN using Radio Interferometry June 2023 – Present
Graduate research, in collaboration with the US Naval Research Lab *Fairfax, VA*

- VLBA data reduction in **CASA**, imaging, and analysis of radio-loud quasar observations at 2 GHz and 8 GHz to investigate varstrometry+radio interferometry as a viable systematic detection method for multi-AGN.

Large VLBA BEAM-ME Jan. 2020 – May 2023
Boston University Institute of Astrophysical Research *Boston, MA*

- VLBA data reduction in **Difmap** of monthly blazar observations to monitor changing brightness and polarizations in blazar jets and their shockwaves and model jet activity over time.
- Determined and compared significant blazar features to investigate correlations between radio jet activity and gamma-ray events.

Patatrack Pixel Tracking Jan. 2022 – Aug. 2022
European Council for Nuclear Research (CERN) *Geneva, Switzerland*

- Tested Patatrack's 4 portability libraries of track reconstruction algorithms across 3 GPU devices on a varying amount of threads to evaluate performance, using CUDA in C++ and Python.

SOFIA 3-Micron Dust Emission Studies June 2020 – Aug. 2020
NASA Goddard Space Flight Center *Greenbelt, MD*

- Fitted 3-5 micron wavelength emission spectra in Python from polycyclic aromatic hydrocarbons surrounding a sample of planetary nebulae to study the relationship between hydrocarbon features and planetary nebulae ages.

Proposals & Grants

Tracing the Origin of AGN Emission in Variable Radio-Quiet Quasars Semester 2026A

- Principal investigator, awarded 24.5 hours at 1, 6, and 8 GHz on the VLBA.

VLBA Characterization of Astrometrically-Variable Quasars Semester 2024B

- Principal investigator, awarded 24 hours at 1, 6, and 15 GHz on the VLBA.

Undergraduate Research Opportunities Program Grant June 2022 – Aug. 2022

- > $**4000** in funding to extend internship at CERN.

Publications

Schwartzman, E., **Fudolig, P.**, et al. Varstrometry for Dual AGN using Radio Interferometry: VaDAR with the VLBA, 2025, ApJ, 987, 200

Conferences, Colloquia, & Presentations

NRAO Socorro Lunch Talks — Jan. 21, 2026
Talk title: A Tale of Two Variabilities: VaDAR and the Brightening J0822+4553 — *Socorro, NM*

DC Astrophysics Graduate Student Conference — Aug. 14-15, 2025
Talk title: Tracing the Origins of Radio Emission in Radio-Quiet Quasars — *Washington, D.C.*

245th American Astronomical Society Meeting — Jan. 12-16, 2025
Talk title: A VLBA View of VaDAR — *National Harbor, MD*

11th Annual Mid-Atlantic Radio-Loud AGN Meeting — November 8, 2024
Talk title: A VLBA View of VaDAR — *Baltimore, MD*

DC Astrophysics Graduate Student Conference — Aug. 15-16, 2024
Talk title: A VLBA View of VaDAR — *Washington, D.C.*

10th Annual Mid-Atlantic Radio-Loud AGN Meeting — October 12, 2023
Talk title: VaDAR: Varstrometry for Dual AGN using Radio Interferometry — *Fairfax, VA*

Boston University Undergraduate Research Symposium — Oct. 21, 2022
Poster title: Pixel-Tracking in CERN's Compact Muon Solenoid — *Boston, MA*

NASA Goddard Summer Poster Session — August 4, 2020
Talk title: Comparing Fits of PAH Emission — *Greenbelt, MD*

Service

SPECTRUM Leader — March 2024 – Present
Organization dedicated to empowerment and community building in STEM students — *Fairfax, VA*

- **Created** and implemented a series of lunch workshops promoting the professional development of graduate and undergraduate students within the GMU College of Science.
- **Created** and implemented a mentorship program for incoming graduate students, including developing mentor/mentee applications, matching mentoring pairs, advertising the opportunity, and organizing related social events. Resulted in a 100% retention rate upon its first implementation in Spring 2024.

Student Advisor — August 2025 – Present
GMU College of Science Dean's Student Advisory Board — *Fairfax, VA*

- **Shared** insights and feedback on the student experience, representing graduate student needs and interests.
- **Collaborated** with the Dean's office on college-level initiatives and improvements.
- **Fostered** a stronger sense of community across programs and disciplines within the College of Science.

Physics Department Representative — August 2024 – May 2025
GMU's Graduate and Professional Student Association (GAPSA) — *Fairfax, VA*

- **Advocated** for fellow Physics graduate students on topics including graduate tuition and fees, fellowship opportunities, student life, and campus events.
- **Collaborated** with other graduate student representatives from all departments on University-level initiatives.
- **Communicated** important information on departmental changes and solicited feedback for upcoming votes.

Peer Mentor — Sep. 2021 – May 2023
BU's PeeRs for Incoming Student Mentorship (PRISM) — *Boston, MA*

- **Guided** first-year students by providing departmental and campus resources, creating four-year plans, sharing personal experiences, and providing support throughout the academic year.

Founder of BU Chapter of Society of Women in Space Exploration — Aug. 2020 – May 2023
Organization supporting the inclusion of women and minorities interested in space-related fields — *Boston, MA*

- As Founder: **authored** organization's constitution and bylaws, assembled an executive board, recruited a faculty advisor, and advertised on campus to build membership.
- As President: **led** biweekly general member meetings, involving devising group activities, planning on-campus events, and networking in the Boston area for collaborations. Also led executive board meetings, delegating tasks and overseeing management of responsibilities.

Related Experience

Graduate Teaching Assistant Aug. 2023 – Present

- Led undergraduate labs in Physics I/II and Astronomy I at George Mason University.

Astronomy on Tap Speaker November 18, 2025

- Presented "Hunting for Dual AGN with VaDAR" to the general public.

Observation Time on Perkins Telescope March 8-10, 2023

- Three nights operating Boston University's Perkins Telescope at Anderson Mesa Observatory in Flagstaff, AZ.

Scientific & Technical Skills

Programming: Python, C++, CUDA, ROOT, Git, LaTeX, Ubuntu, Jupyter Notebook, Mathematica, Bash/Shell
Astronomical: NRAO CASA, NRAO AIPS, Caltech Difmap, PyBDSF, TopCat, SAOImage DS9, Maxim DL
World Languages: English (native), Spanish (advanced), French (intermediate), Tagalog (basic)